

CPIT-240 FINAL (Spring 2024)

Answer TRUE or FALSE:

1. Normalization is used to increase redundancy and dependency of data
2. Functional dependencies (FD) are a property of the attributes in the schema
3. We can be sure that a functional dependency does not exist just by looking at the database state
4. Aggregate functions can be used in the SELECT clause or in HAVING clause
5. Aggregate functions used to summarize information from a single tuple into a multiple tuple
6. SQL does not automatically eliminate duplicate tuples in query results
7. SQL is a programming language designed for managing and querying databases
8. SQL allows users to retrieve, manipulate, and manage data stored in a database
9. SQL commands are not case-sensitive
10. SQL supports both data definition and data manipulate operations
11. The relation state is a subset of the Cartesian product of the domains of its attributes
12. FULL OUTER JOIN combines result if LEFT and RIGHT OUTER JOIN
13. The result of the operation $R \cap S$, is a relation that includes all tuples that are either in R or in S or in both R and S
14. The UNION operation, the two operands must be type compatible
15. Can the Division operation be used in Relational Algebra to find common elements between different tables? **(This one has 1 mark and the rest have 0.5 mark for each)**

Fill the blank with the operation

1. _____ is used to extract rows in the first table that do not exist in the second table
 2. _____ is used to extract the specified columns from the table
 3. _____ is used to extract common rows between two tables
 4. _____ is used to extract rows that meet a certain condition
 5. _____ is used to combines rows from multiple tables based on a specified condition
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Based on the t in the following table, check all functional dependencies (for primary and not primary attributes)? Why?

ST_NO	ST_Name	ST_State	ST_Country	ST_BOD
1	Ali	Kerala	India	14 / 11 / 2005
2	Omar	Makkah	KSA	01 / 12 / 2004
3	Khalid	Kerala	India	20 / 01 / 2005
4	Ali	Makkah	KSA	14 / 11 / 2005

Examine the table shown below

<u>branchNo</u>	branchAddress	telNo
B1	City Center, Jeddah, JA910	012 654 321, 012 546 231
B2	Avenue, Jeddah, JA800	012 321 654, 012 123 456, 012 213 546
B3	City Center, Jeddah, JA910	012 123 456
B4	Avenue, Jeddah, JA800	012 456 123

Why is the above table not in 1NF?

Describe and illustrate the process of normalizing the data shown in the table to third normal form (3NF), and identify the primary, foreign keys in your 3NF relations

Examine the table shown below

StaffNo	branchNo	branchAddress	name	position	hoursPerWeek
455	B2	City Center, Jeddah, JA910	Ellen	Assistant	16
455	B4	Avenue, Jeddah, JA800	Ellen	Assistant	9
460	B2	City Center, Jeddah, JA910	Dave	Assistant	14
460	B4	Avenue, Jeddah, JA800	Dave	Assistant	10

Why is the above table not in 2NF?

Describe and illustrate the process of normalizing the data shown in the table to third normal form (3NF), and identify the primary, foreign keys in your 3NF relations

Examine the table shown below

<u>branchNo</u>	branchAddress	telNo	mgrStaffNo	name
B1	City Center, Jeddah, JA910	012 654 321	10	Ellen
B2	Avenue, Jeddah, JA800	012 321 654	11	Mary
B3	City Center, Jeddah, JA910	012 123 456	12	Sally
B4	Avenue, Jeddah, JA800	012 456 123	13	Peters

Why is this table not in 3NF?

Describe and illustrate the process of normalizing the data shown in the table to third normal form (3NF), and identify the primary, foreign keys in your 3NF relations

Write the Structure of SQL Queries

We have the following database, answer the queries mentioned below:

Students

<u>Student_id</u>	Student_Name	department	Grade
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Student_grades

<u>Student_id</u>	Course_id	Weight	Grade
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Exam_grades

Student_id	Mid_grade	assignment1_grade	assignment2_grade	final_exam_grade	Total
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How many students are registered in all departments? (in SQL & Relational Algebra)

What are the names of the students who scored above 90 on the second assignment? (in SQL & Relational Algebra)

What is the average score of students on the final exam? (in SQL)

What are the final grades of all students in descending order? (in SQL & Relational Algebra)

Retrieve the department name and the count of the number of students belonging to each department? (in SQL & Relational Algebra)

What departments do the students with the highest scores in the final exam study in? (in SQL)

What are the students who did not pass the final exam? [Hint: Student who did not pass have their names not found in the exam_grades table] (in SQL)